9. Triangular Matrix

A square matrix in which all the entries above the leading diagonal are zero is called a lower triangular matrix.

If all the entries below the leading diagonal are zero, then it is called an upper triangular matrix.

Definition : A square matrix $A = (a_{ij})_{n\times n}$ is called upper triangular matrix if $a_{ij} = 0$ for $i > j$ and is called lower triangular matrix if $a_{ij} = 0$, $i < j$.

For example, $A = \begin{pmatrix} 1 & 7 & -3 \\ 0 & 2 & 4 \\ 0 & 0 & 7 \end{pmatrix}$ is an upper triangular matrix and $B = \begin{pmatrix} 8 & 0 & 0 \\ 4 & 5 & 0 \\ -11 & 3 & 1 \end{pmatrix}$ is a lower triangular matrix.

Equal Matrices

Two matrices A and B are said to be equal if and only if they have the same order and each element of matrix A is equal to the corresponding element of matrix B. That is, $a_{ij} = b_{ij}$ for all i, j.

For example, if $A = \begin{pmatrix} 5 & 1 \\ 0 & 3 \end{pmatrix}$,

$B = \begin{pmatrix} 1^2 + 2^2 & \sin^2\theta + \cos^2\theta \\ 1 + \frac{3}{2} - \frac{5}{2} & 2 + \sec^2\theta - \tan^2\theta \end{pmatrix}$ then we note that A and B have same order and $a_{ij} = b_{ij}$ for every i, j. Hence A and B are equal matrices.

Progress Check

1. The number of column(s) in a column matrix are _____.
2. The number of row(s) in a row matrix are _____.
3. The non-diagonal elements in any unit matrix are _____.
4. Does there exist a square matrix with 32 elements?

The negative of a matrix

The negative of a matrix $A_{m\times n}$ denoted by $-A_{m\times n}$ is the matrix formed by replacing each element in the matrix $A_{m\times n}$ with its additive inverse.

Additive inverse of an element k is $-k$. That is, every element of $-A$ is the negative of the corresponding element of A.

For example, if $A = \begin{pmatrix} 2 & -4 & 9 \\ 5 & -3 & -1 \end{pmatrix}_{2\times 3}$ then $-A = \begin{pmatrix} -2 & 4 & -9 \\ -5 & 3 & 1 \end{pmatrix}_{2\times 3}$

Example 3.56 Consider the following information regarding the number of men and women workers in three factories I, II and III.

Factory	Men	Women
I	23	18
II	47	36
III	15	16

Represent the above information in the form of a matrix. What does the entry in the second row and first column represent?

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